



Predict Both

Quick facilitation notes & answer key

Goal

Students PREDICT the combined outcome of two concurrent scripts before running them — the C3.2 / 'predict what happens when both start' skill. Predicting both landings forces them to trace two parallel scripts in their head at once, then verify against the run. Here Bit lands on 3 and Pixel on 2 (they do not meet).

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: have students write their prediction on a sticky note, then move two counters (green Bit, purple Pixel) on a 0-5 path to check.

Run it (about 20 min)

1. I DO: model predicting Bit ONLY — 'start 0, forward 2, forward 1 — I predict 3.' Write the prediction down BEFORE moving.
2. WE DO: predict Pixel together — 'start 5, back 2, back 1 — I predict 2.' Then press the green flag and run both.
3. YOU DO: students compare their predictions (Bit 3, Pixel 2) to the run, and answer whether they meet (NO).
4. Connect: 'Predicting two scripts at once is reading concurrent code in your head before the computer runs it.'

Answer key (modelled by the run-gate)

Ex 1 (predict) — Bit: 0 → forward 2 → 2 → forward 1 → 3 (predict 3). Pixel: 5 → back 2 → 3 → back 1 → 2 (predict 2).

Ex 2 (run) — the run confirms Bit on 3 and Pixel on 2; matching predictions.

Ex 3 — 3 and 2 are different numbers → NO, they do not meet (they land right next to each other).

Interaction note: both scripts run at the same time from the green flag; a correct prediction names BOTH final positions.

Watch for & differentiate

Common error: predicting only one sprite — stress that the prediction must cover BOTH because both run at once.

Common error: changing the prediction after seeing the answer — have them commit the prediction in pen first.

Simplify: predict Bit fully, run it, then predict Pixel, run it — before combining.

Extend: ask 'whose prediction was harder, and why?' (Pixel moves backward, which some find trickier).

Success indicator: the child predicts Bit 3 and Pixel 2, runs to confirm, and says they do not meet.